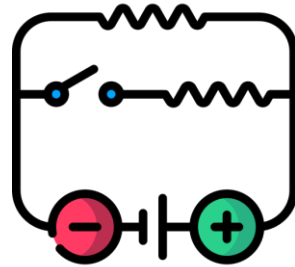


EMJ12103

ELECTRIC CIRCUIT 1



Chapter 4: Mesh Analysis

DR. AZRALMUKMIN AZMI

Mesh Analysis

- ▶ Similar to nodal analysis.
- ▶ Developed based on the systematic approach to KVL to find all circuit variables without having to sacrifice any of the elements.
- ▶ General procedure which is making use of mesh current in circuit analysis as key solutions.



Mesh Analysis - Terms

- ▶ A **Mesh** - a loop which does not contain any other loops within it.
- ▶ **Mesh current** – Assigned unknown current flows around the perimeter of the particular mesh/loop.
- ▶ **Element current** – Actual current through any element or branch in the circuit.



Mesh Analysis - Steps

STEP 1: Assign mesh currents $i_1, i_2, \dots, i_n$ to the n meshes.

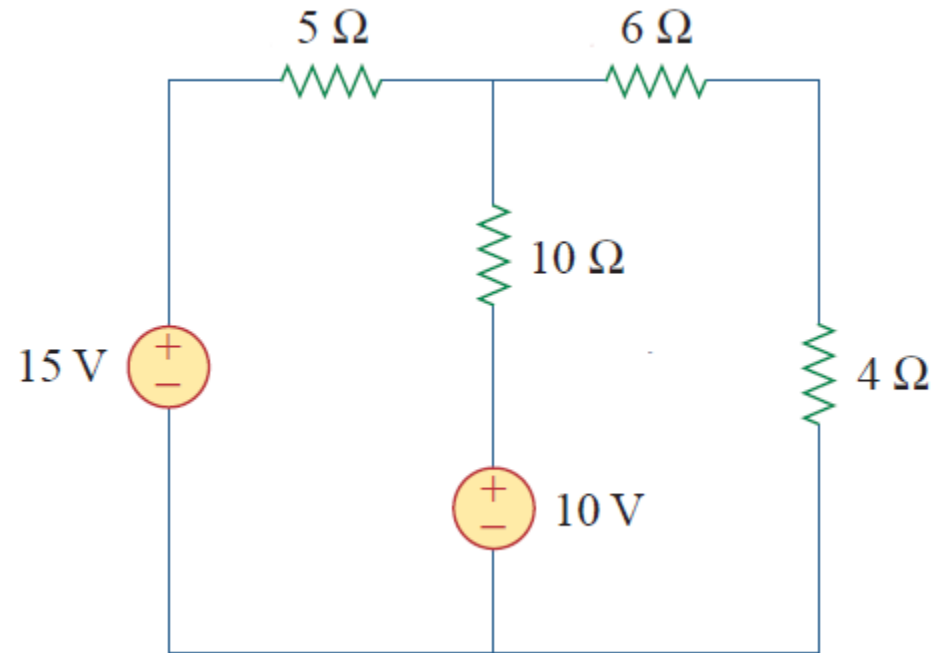
STEP 2: Apply KVL at each of the n meshes. Use Ohm's Law to express the voltage in term of the mesh currents.

STEP 3: Solve the resulting simultaneous equation to get the mesh currents.



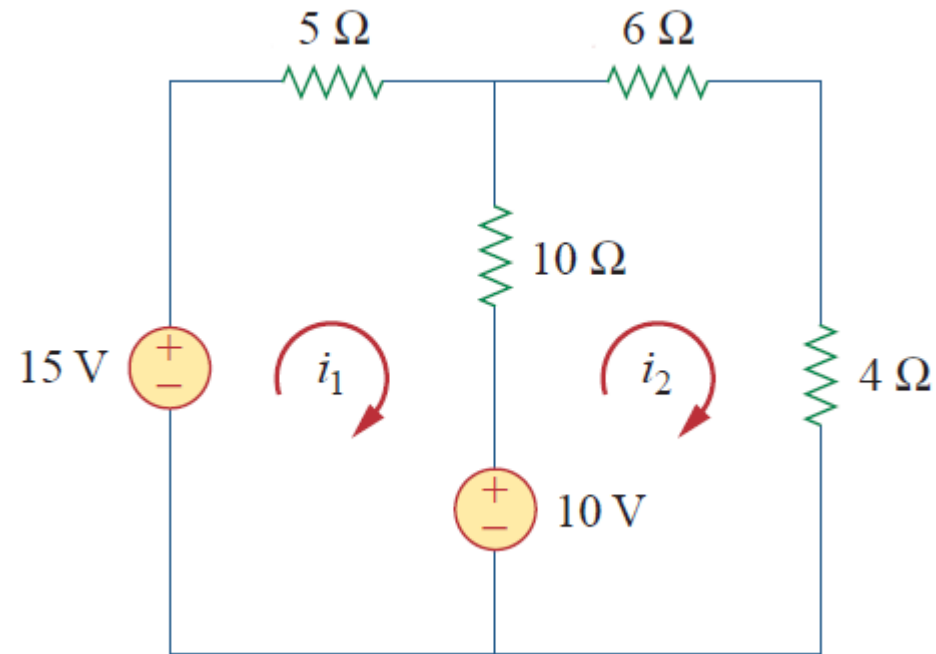
Mesh Analysis (Cont...)

- Apply mesh analysis to the network:



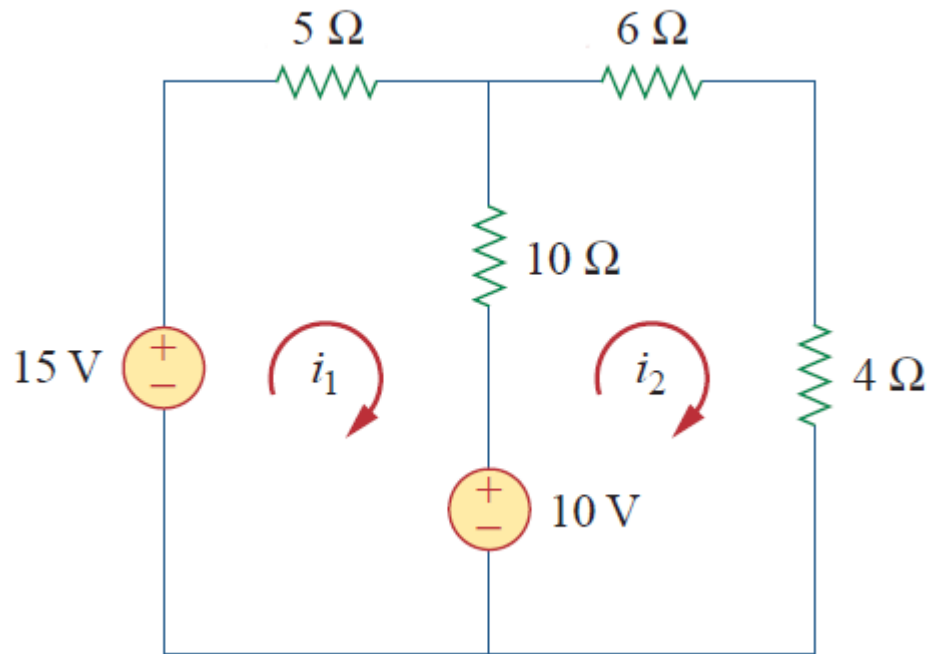
Mesh Analysis (Cont...)

- **STEP 1:** Assign mesh currents $i_1, i_2, \dots, i_n$ to the n meshes.



Mesh Analysis (Cont...)

STEP 2: Apply KVL at each of the n meshes. Use Ohm's Law to express the voltage in term of the mesh currents.



Using KVL at loop i_1

$$V_{15V} + V_{5\Omega} + V_{10\Omega} + V_{15V} = 0$$

$$-15 + 5(i_1) + 10(i_1 - i_2) + 10 = 0$$

$$15i_1 - 10i_2 = 5$$

Using KVL at loop i_2

$$V_{10V} + V_{10\Omega} + V_{6\Omega} + V_{4\Omega} = 0$$

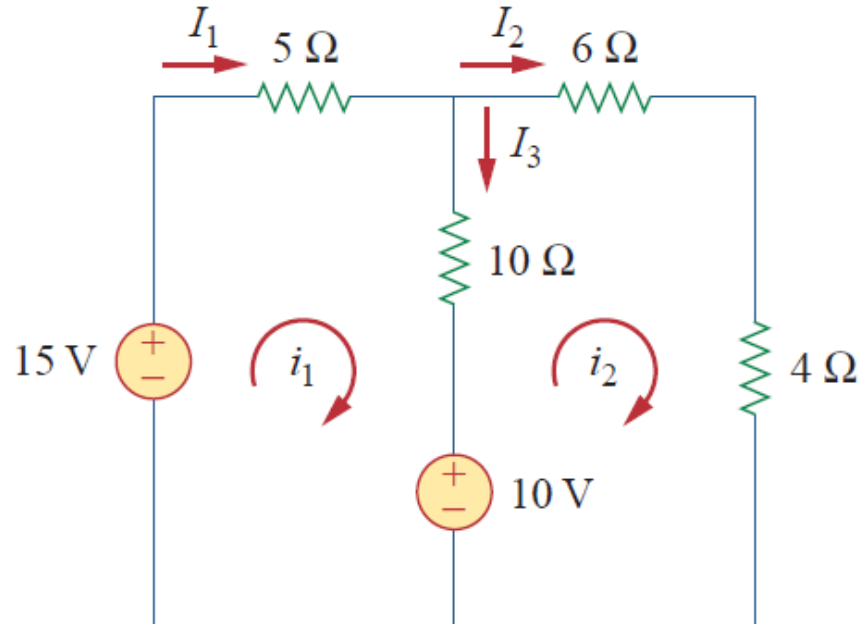
$$-10 + 10(i_2 - i_1) + 6(i_2) + 4(i_2) = 0$$

$$10i_1 - 20i_2 = -10$$

Mesh Analysis (Cont...)

STEP 3: Solve the resulting simultaneous equation to get the mesh currents.

Solve i_1 and i_2 using substitution method



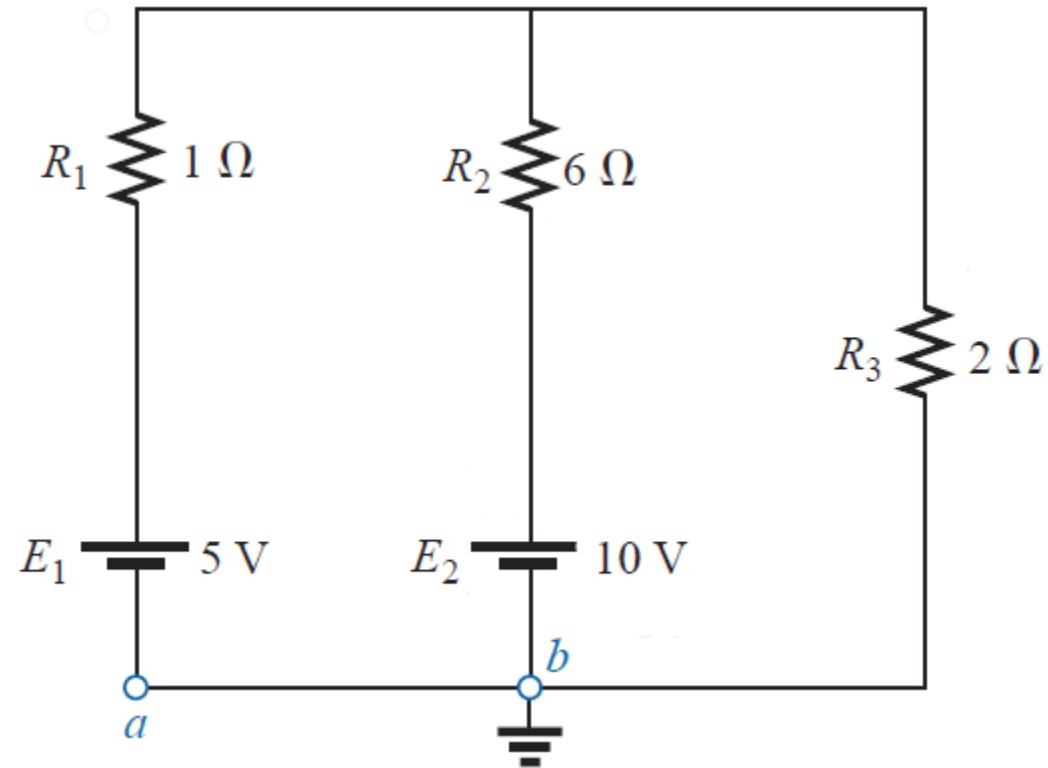
$$I_1 = i_1 = 1 \text{ A}$$

$$I_2 = i_2 = 1 \text{ A}$$

$$I_3 = i_1 - i_2 = 0 \text{ A}$$

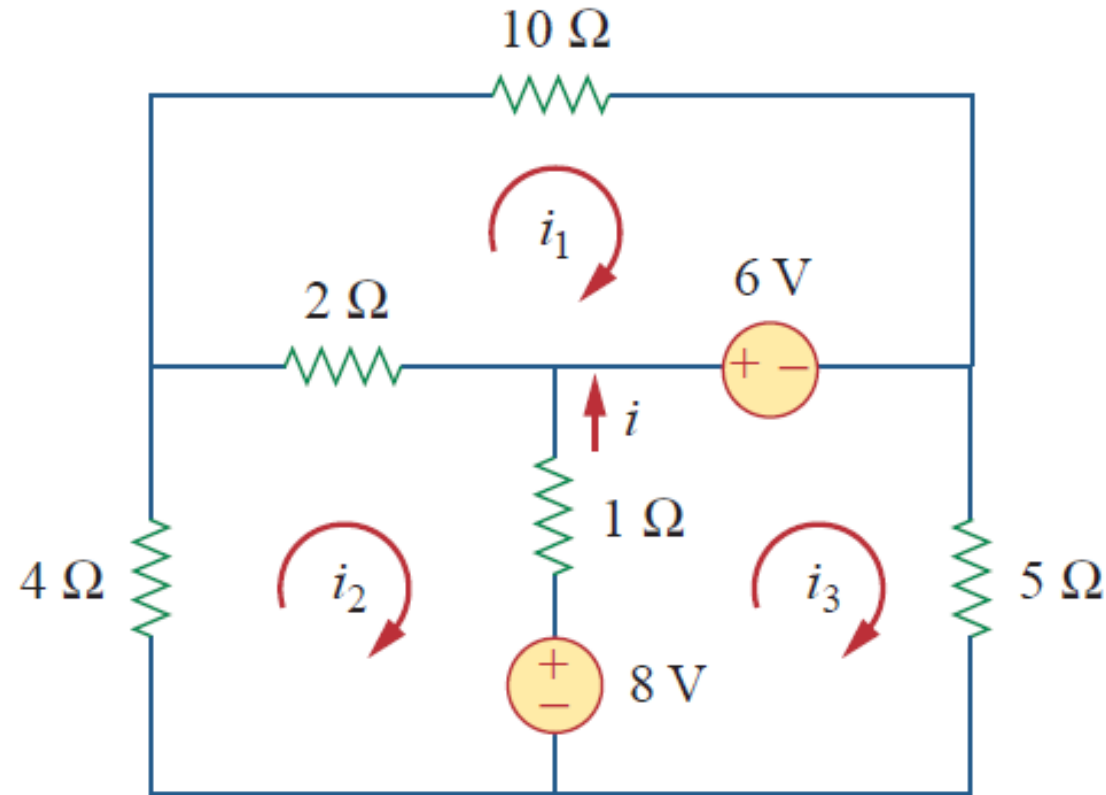
Mesh Analysis (Cont...)

- Apply mesh analysis to the network:



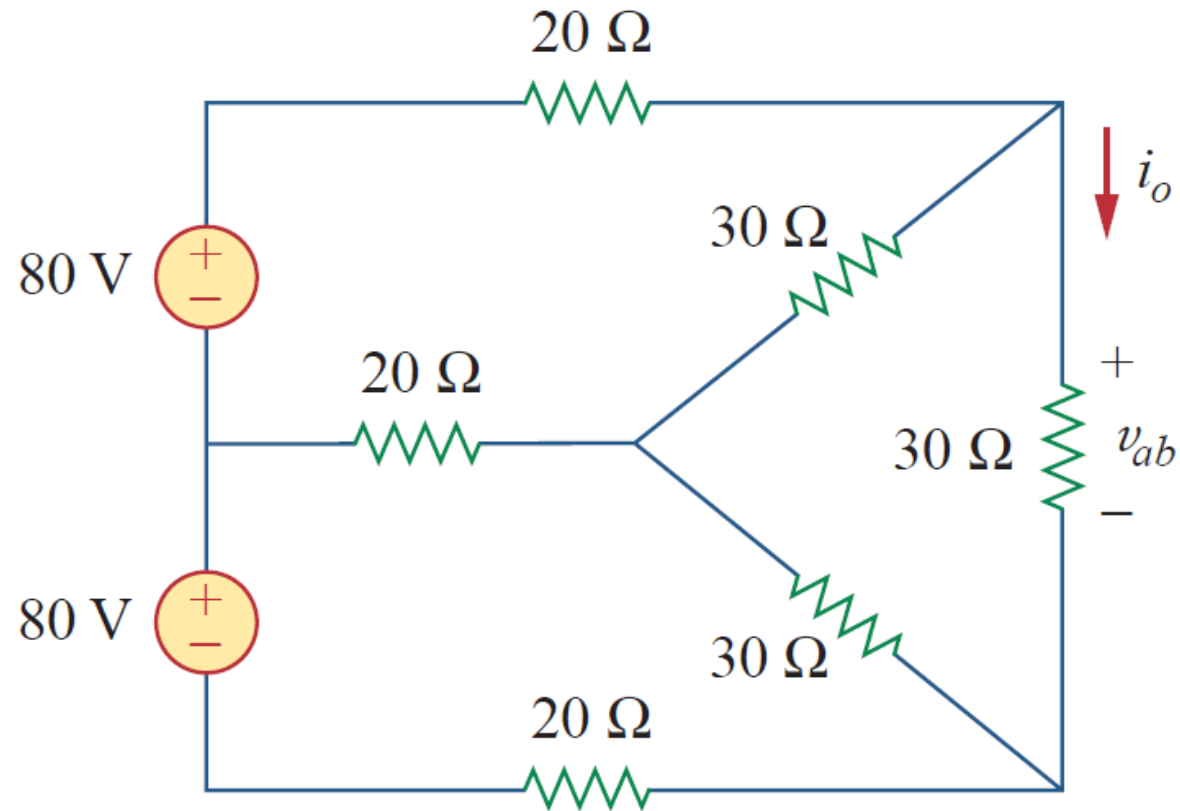
Mesh Analysis (Cont...)

- Apply mesh analysis to find i :



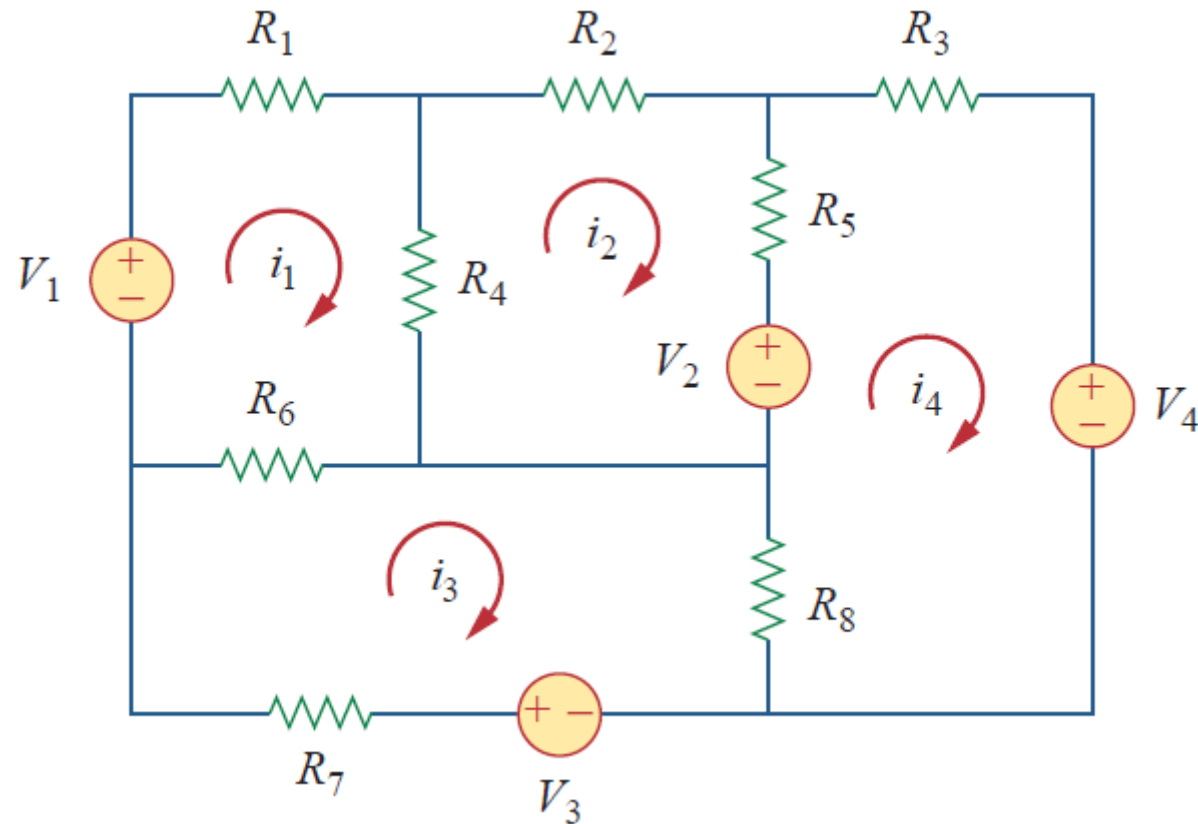
Mesh Analysis (Cont...)

- Apply mesh analysis to find V_{ab} and i_o :



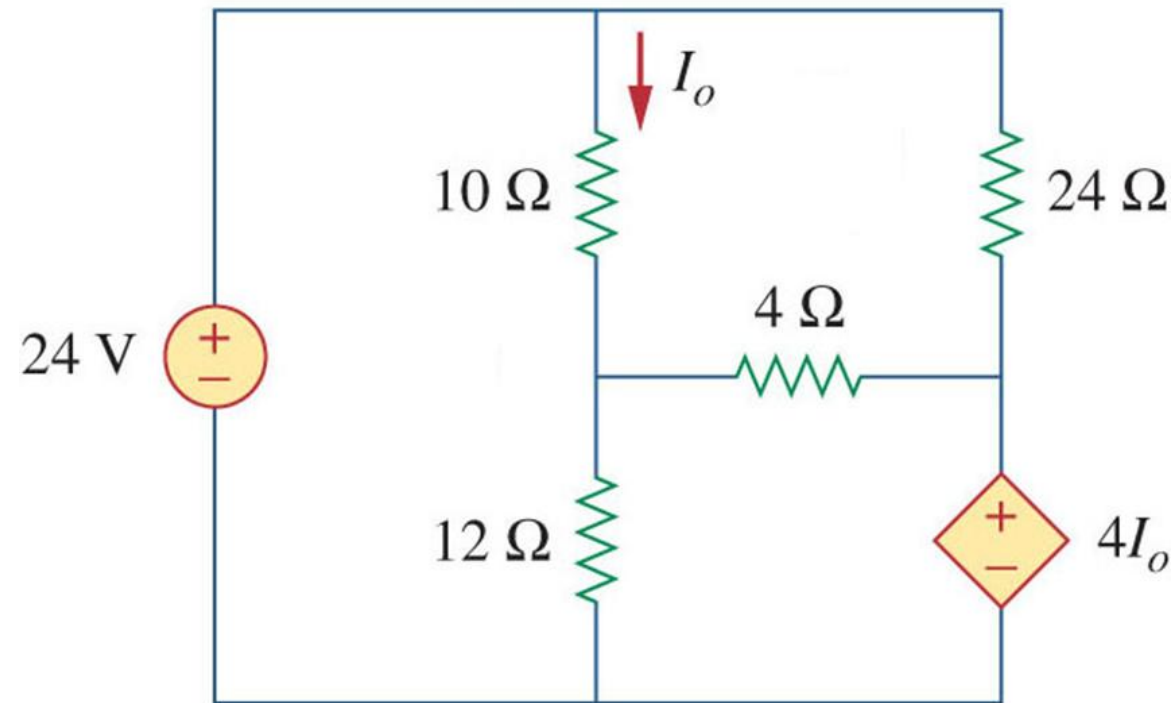
Mesh Analysis (Cont...)

- Write the mesh equation for the network:



Mesh Analysis (Cont...)

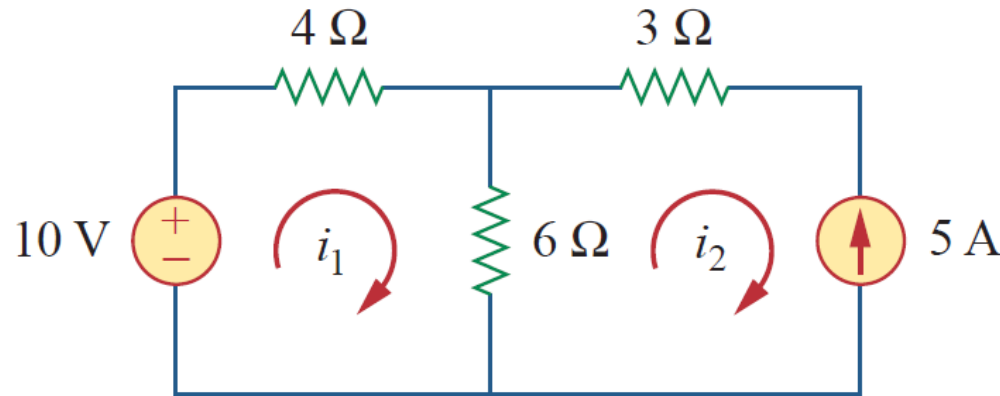
- Find i_o using mesh analysis:



Mesh Analysis with Current Source

Case 1 :

When a current source exists only in one mesh, current of that mesh equal current source.



$$i_2 = -5 A$$

Apply KVL at loop i_1

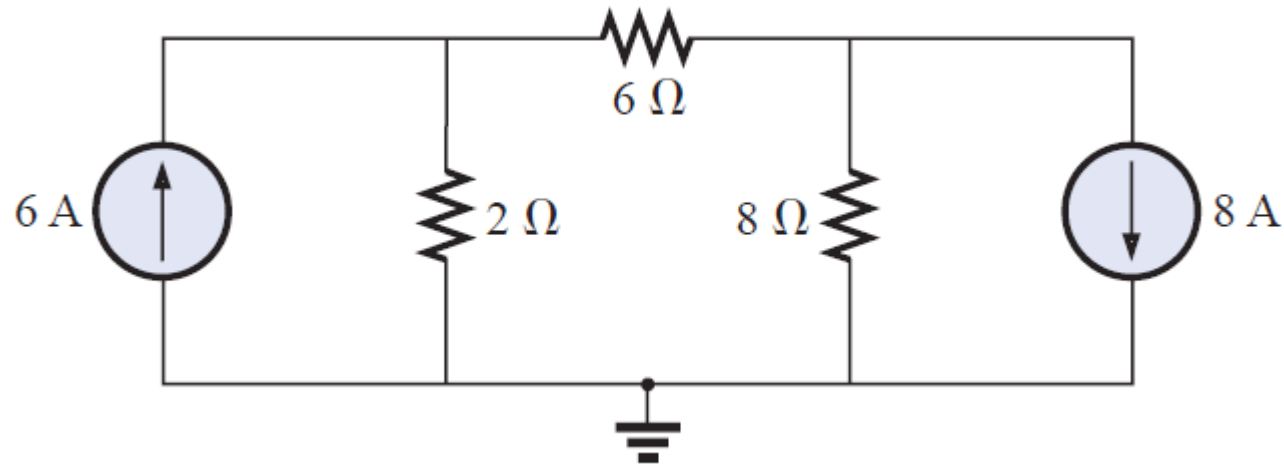
Then

$$-10 + 4i_1 + 6(i_1 - i_2) = 0$$

$$i = -2 A$$

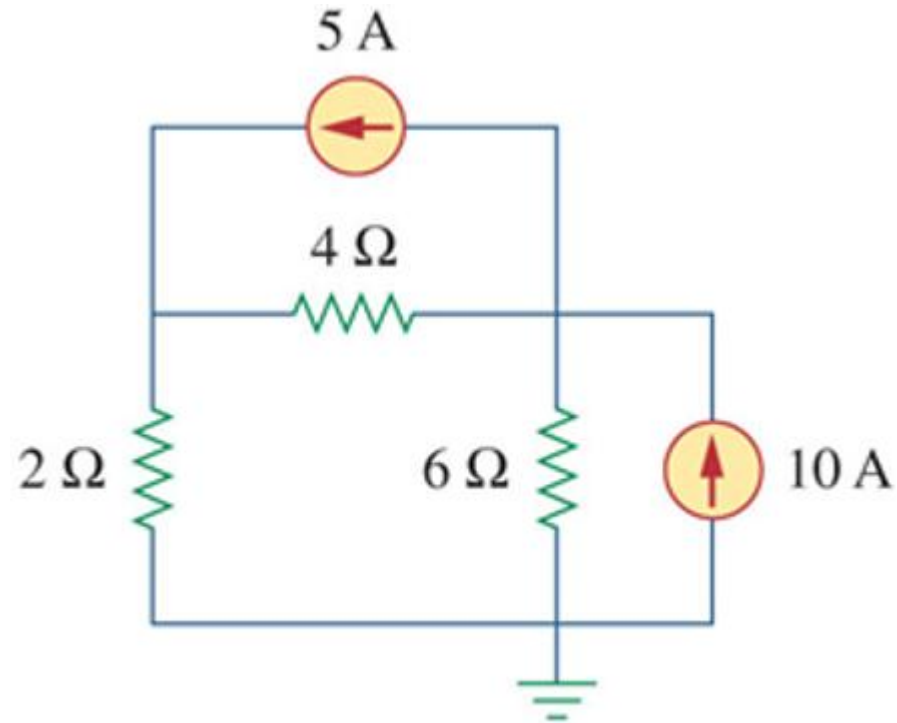
Mesh Analysis with Current Source (Cont...)

- ▶ Using mesh analysis, determine the currents of the network:



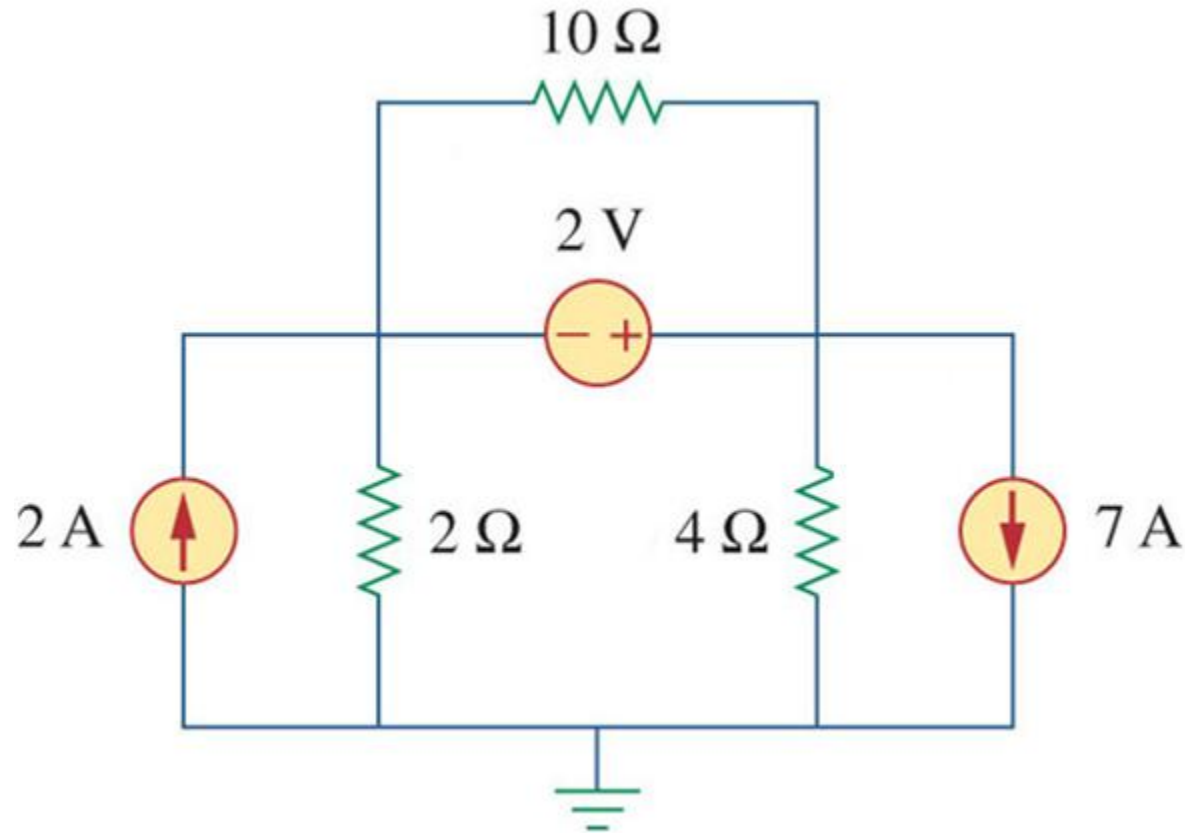
Mesh Analysis with Current Source (Cont...)

- Determine current through $2\ \Omega$ using mesh analysis:



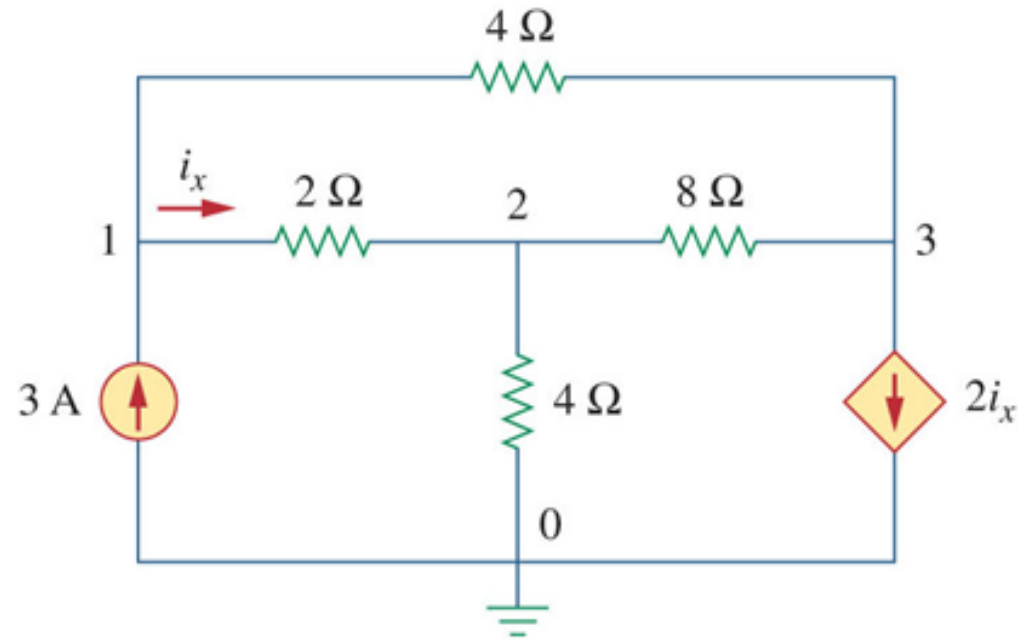
Mesh Analysis with Current Source (Cont...)

- Find branch current using mesh analysis:



Mesh Analysis (Cont...)

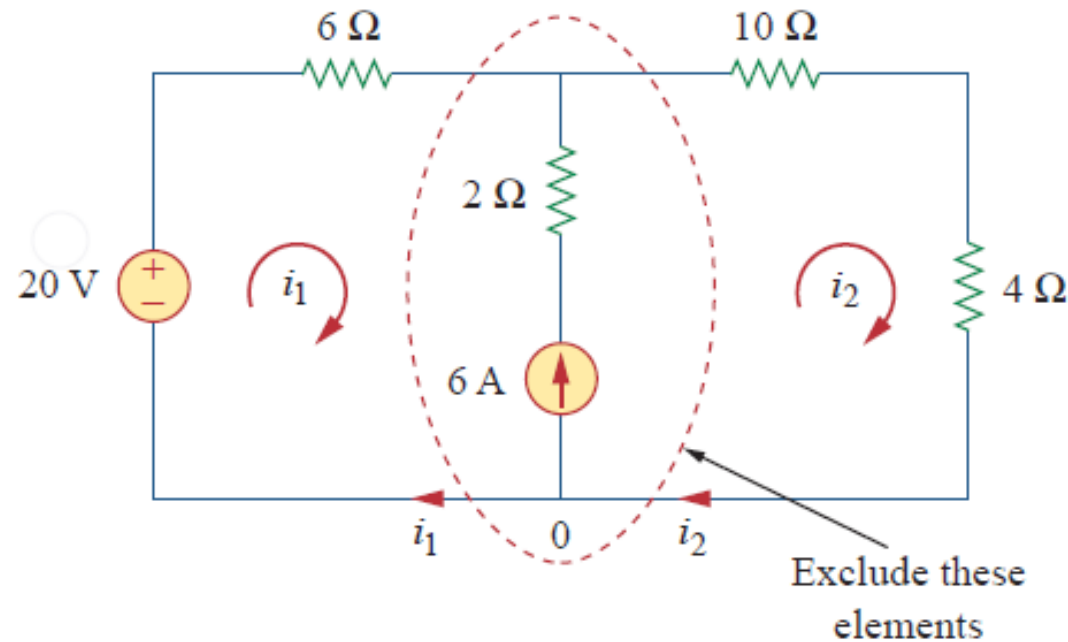
- Find i_x using mesh analysis:



Mesh Analysis with Current Source

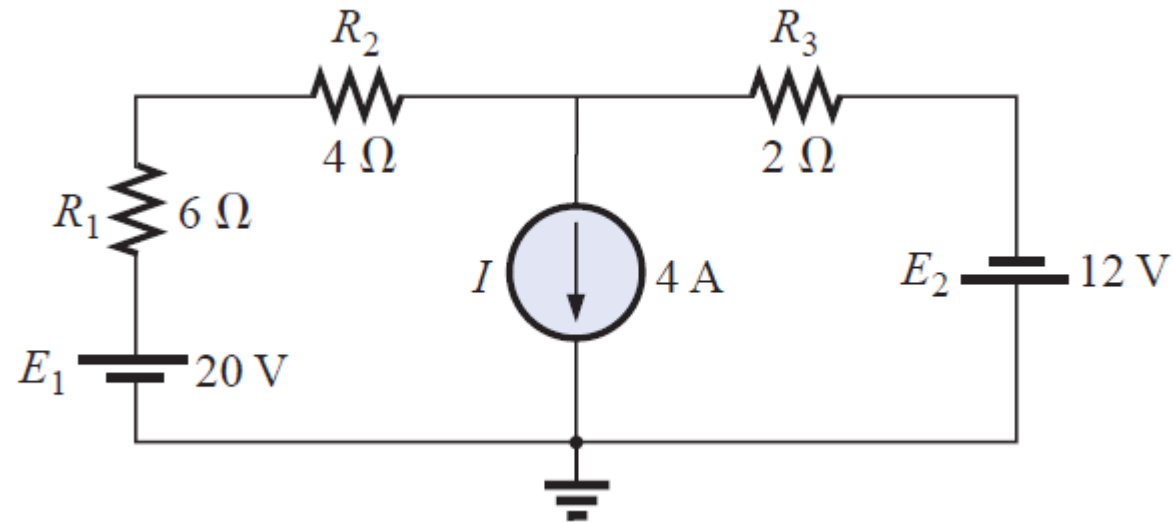
Case 2 :

When a current source (dependent and independent) exists between two meshes, we create a **supermesh** by excluding the current source and any elements connected in series with it.



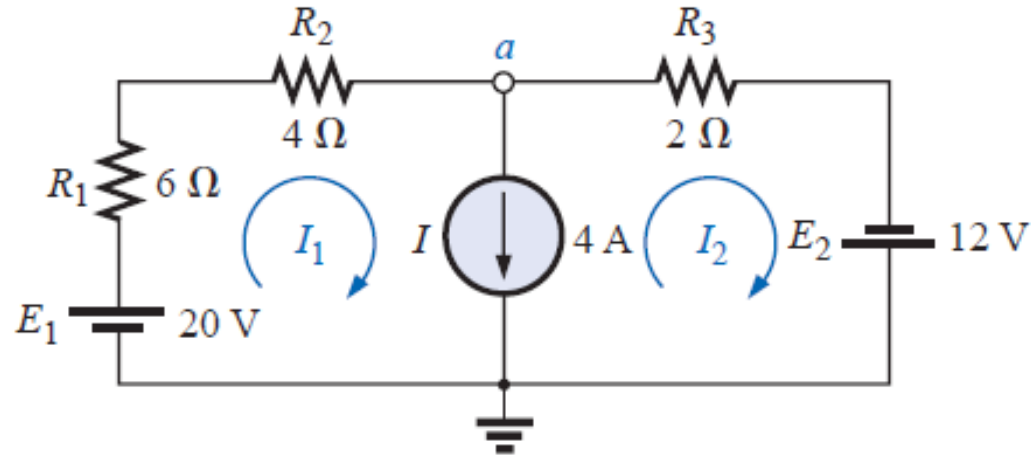
Supermesh

- Using mesh analysis, determine the currents of the network:

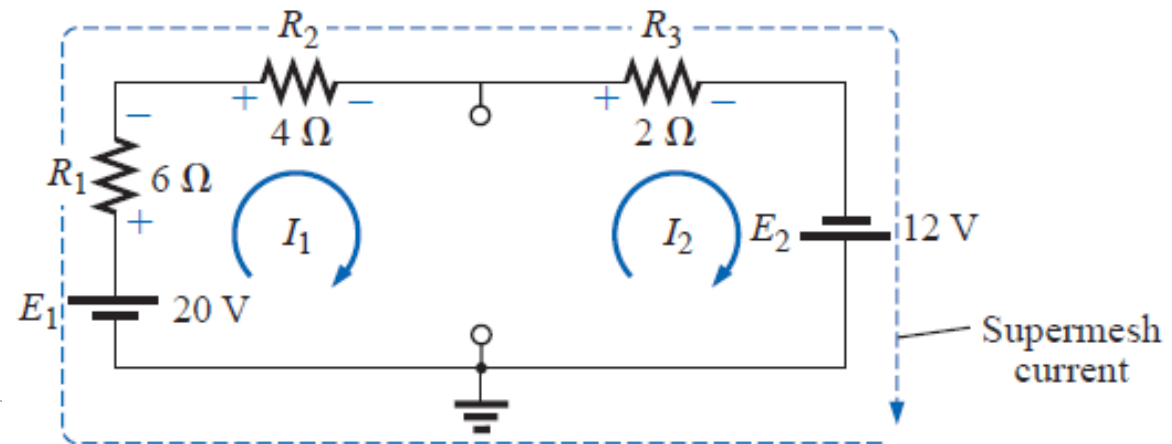


Supermesh (Cont...)

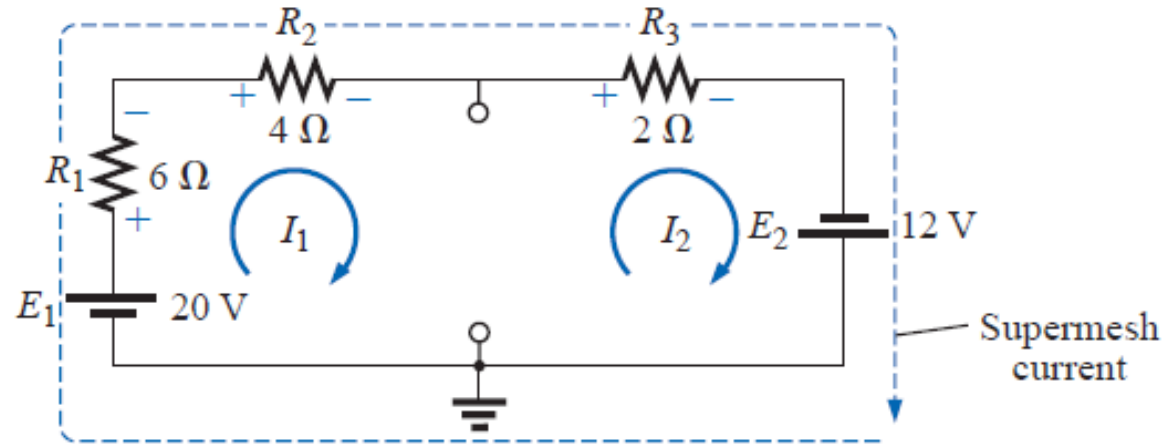
Define mesh current



Define Supermesh current



Supermesh (Cont...)



Apply KVL at supermesh

$$-20 + 6I_1 + 4I_1 + 2I_2 - 12 = 0$$

$$10I_1 + 2I_2 = 32$$

Solve:

$$I_1 = 3.33 \text{ A}$$

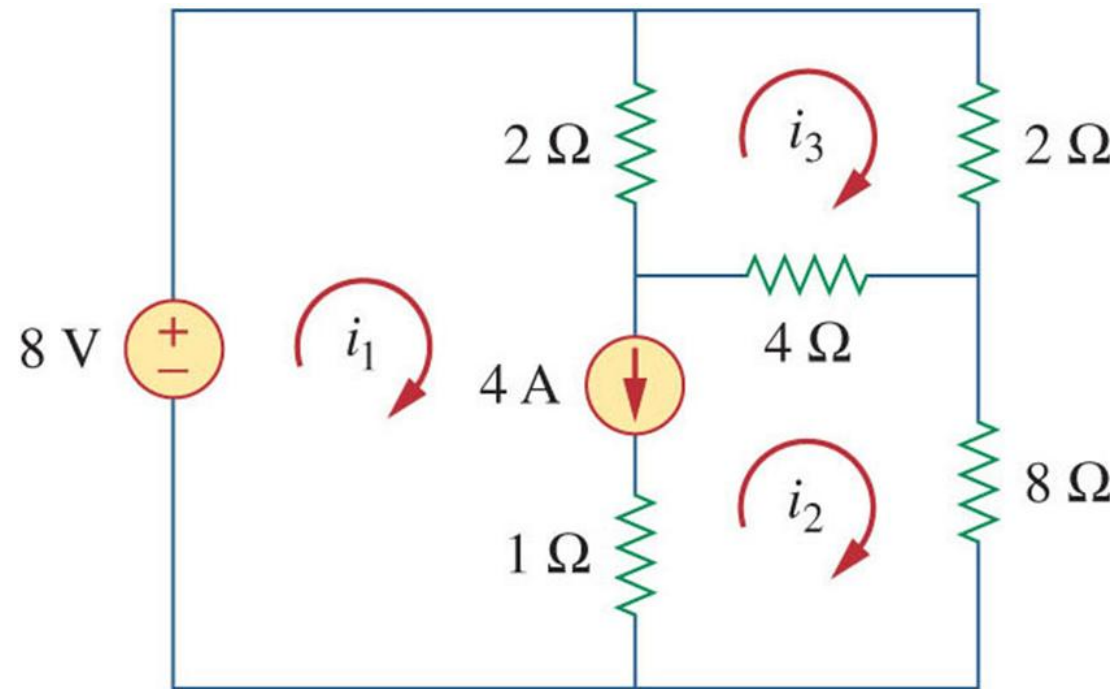
$$I_2 = -0.67 \text{ A}$$

Apply KCL at supermesh

$$I_1 = 4 + I_2$$

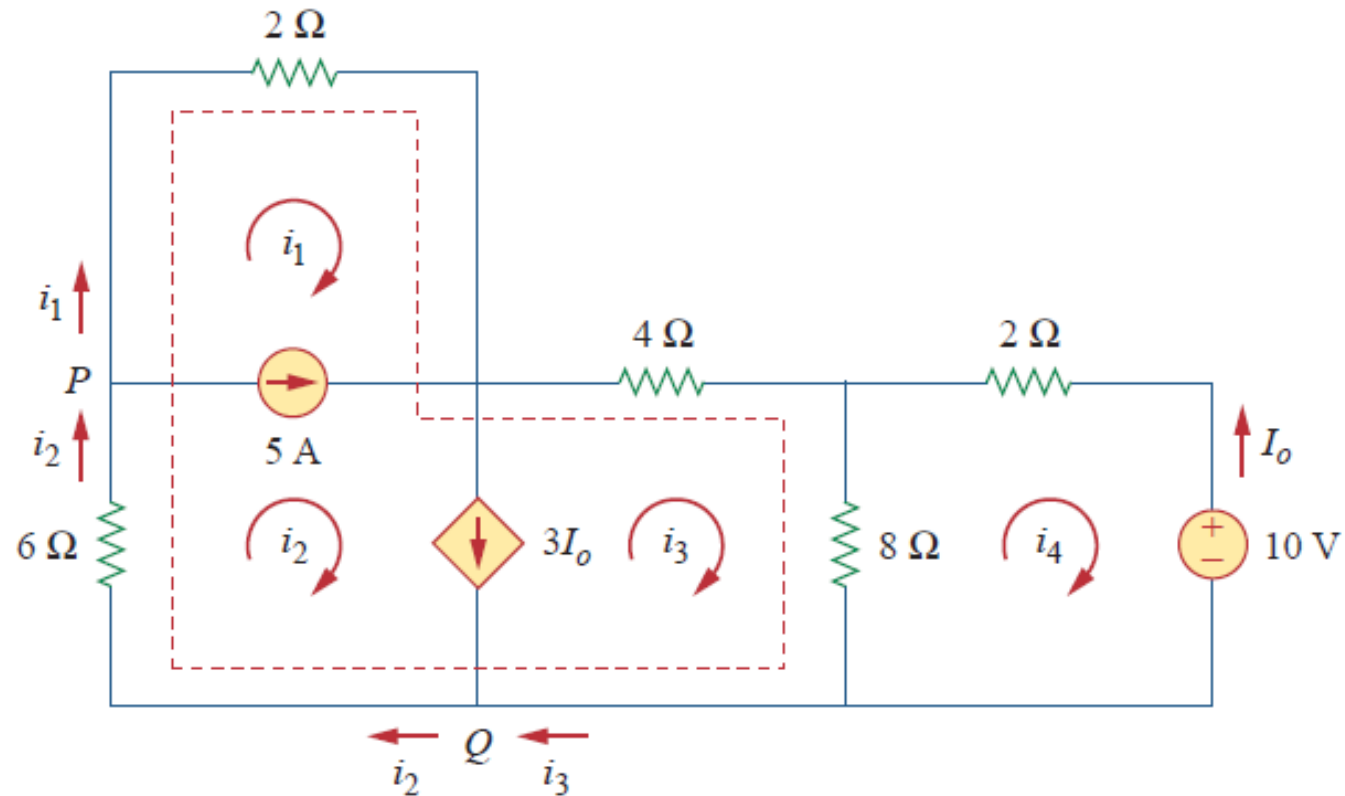
Supermesh (Cont...)

- ▶ Using mesh analysis, determine i_1 , i_2 and i_3 :



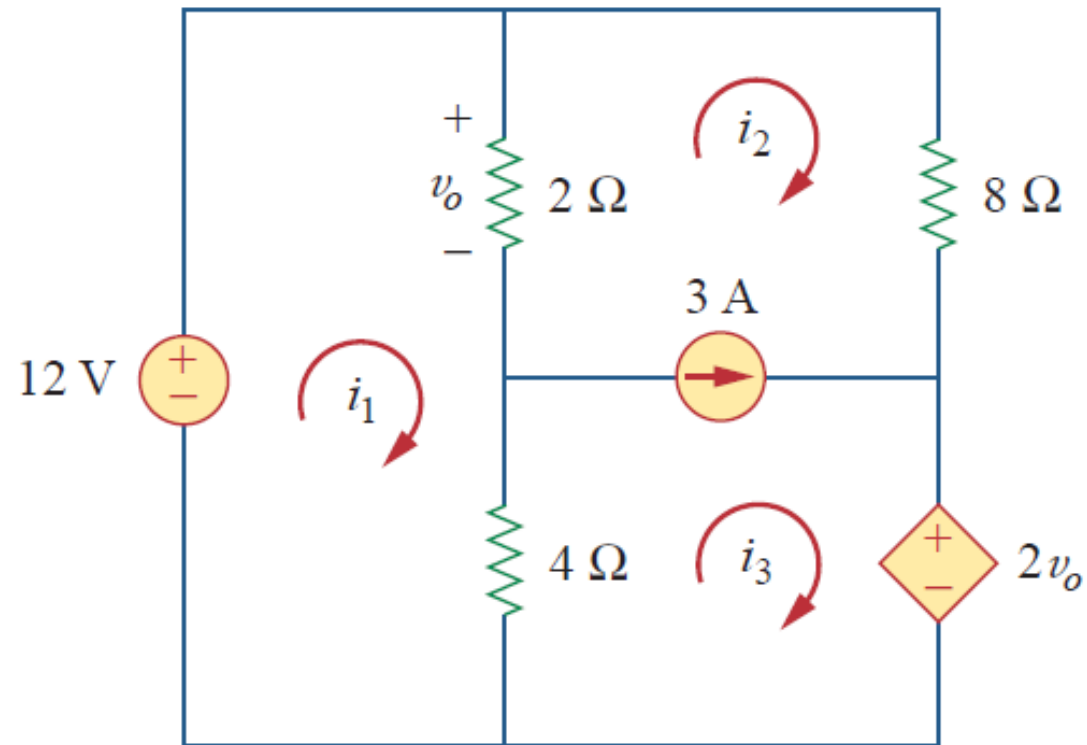
Supermesh (Cont...)

- Using mesh analysis, determine i_1 , i_2 , i_3 and i_4 :



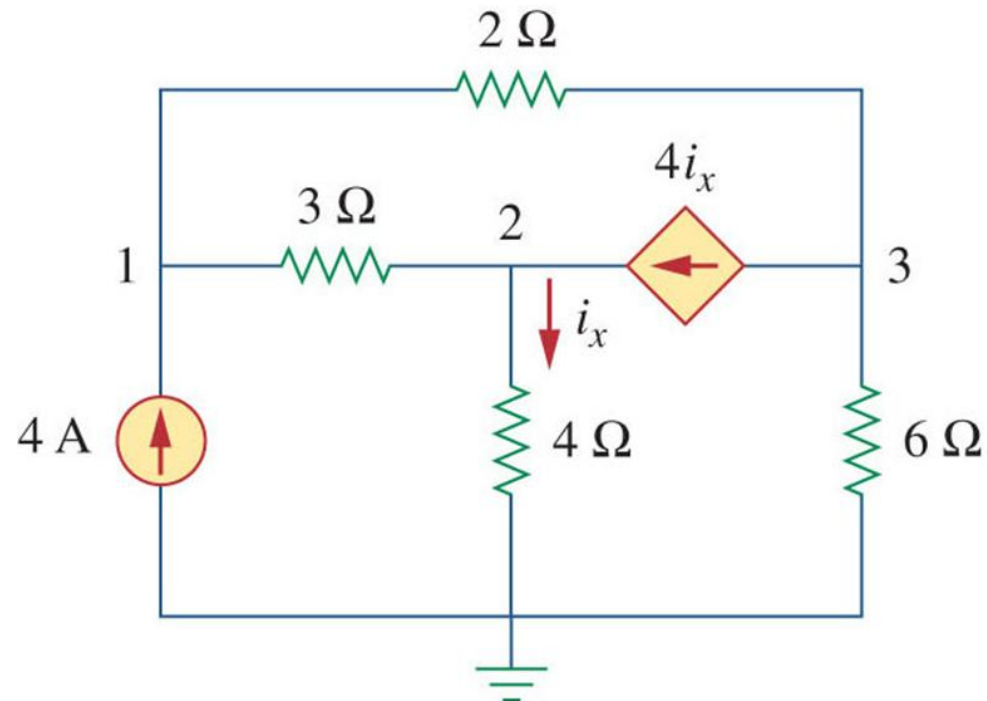
Mesh Analysis (Cont...)

- Find v_o using mesh analysis:



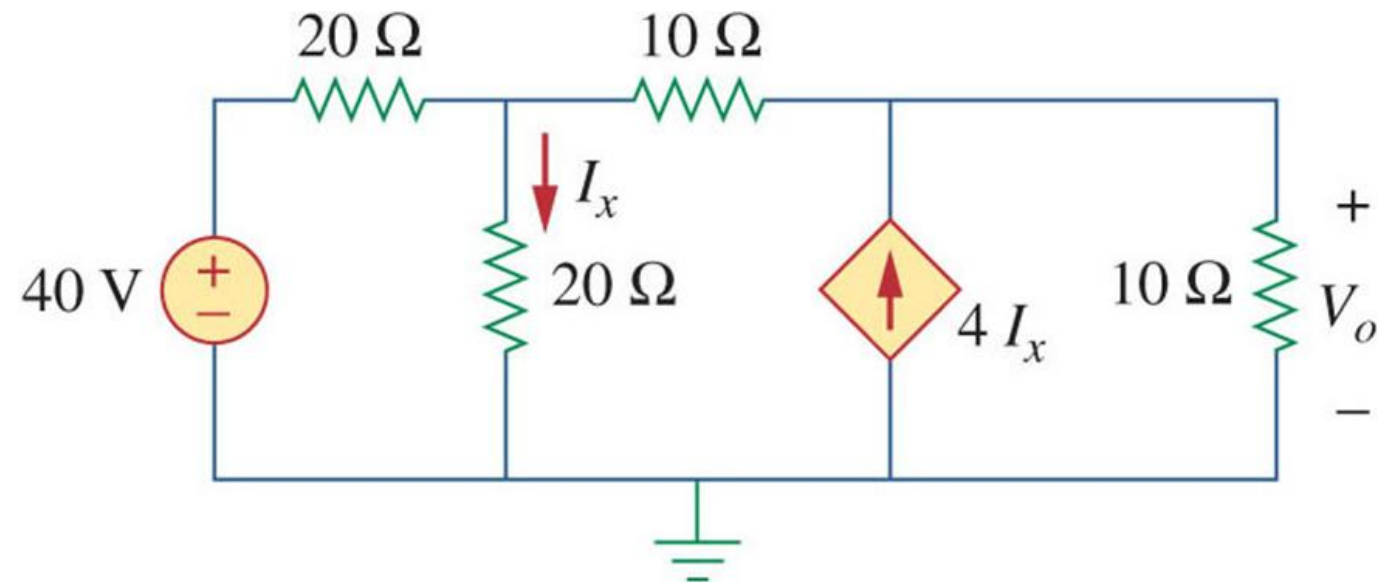
Supermesh (Cont...)

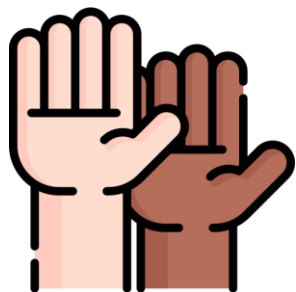
- Using mesh analysis, determine i_x :



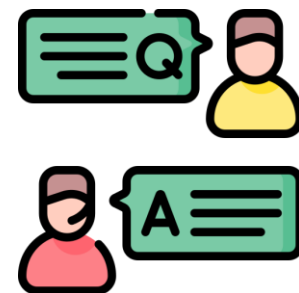
Supermesh (Cont...)

- Determine V_o using mesh analysis:





Q & A



Contact



DR. AZRALMUKMIN AZMI

02-01-A, Block 2, FTKE 2

Fakulti Kejuruteraan & Teknologi Elektrik

Universiti Malaysia Perlis (UniMAP)

azralmukmin@unimap.edu.my

+60193867034

